Research Notes on Machine translation

Machine translation

>> Section 1

Why does a translator need a whole workday to translate five pages, and not an hour or two? ..... About 90% of an average text corresponds to these simple conditions. But unfortunately, there's the other 10%. It's that part that requires six [more] hours of work. There are ambiguities one has to resolve. For instance, the author of the source text, an Australian physician, cited the example of an epidemic which was declared during World War II in a "Japanese prisoners of war camp". Was he talking about an American camp with Japanese prisoners or a Japanese camp with American prisoners? The English has two senses. It's necessary therefore to do research, maybe to the extent of a phone call to Australia.

>> Section 2

In information extraction, named entities, in a narrow sense, refer to concrete or abstract entities in the real world such as people, organizations, companies, and places that have a proper name: George Washington, Chicago, Microsoft. It also refers to expressions of time, space and quantity such as 1 July 2011, $500. In the sentence "Smith is the president of Fabrionix" both Smith and Fabrionix are named entities, and can be further qualified via first name or other information; "president" is not, since Smith could have earlier held another position at Fabrionix, e.g. Vice President. The term rigid designator is what defines these usages for analysis in statistical machine translation. Named entities must first be identified in the text; if not, they may be erroneously translated as common nouns, which would most likely not affect the BLEU rating of the translation but would change the text's human readability. They may be omitted from the output translation, which would also have implications for the text's readability and message. Transliteration includes finding the letters in the target language that most closely correspond to the name in the source language. This, however, has been cited as sometimes worsening the quality of translation. For "Southern California" the first word should be translated directly, while the second word should be transliterated. Machines often transliterate both because they treated them as one entity. Words like these are hard for machine translators, even those with a transliteration component, to process. Use of a "do-not-translate" list, which has the same end goal – transliteration as opposed to translation – still relies on correct identification of named entities. A third approach is a class-based model. Named entities are replaced with a token to represent their "class"; "Ted" and "Erica" would both be replaced with "person" class token. Then the statistical distribution and use of person names, in general, can be analyzed instead of looking at the distributions of "Ted" and "Erica" individually, so that the probability of a given name in a specific language will not affect the assigned probability of a translation. A study by Stanford on improving this area of translation gives the examples that different probabilities will be assigned to "David is going for a walk" and "Ankit is going for a walk" for English as a target language due to the different number of occurrences for each name in the training data. A frustrating outcome of the same study by Stanford (and other attempts to improve named recognition translation) is that many times, a decrease in the BLEU scores for translation will result from the inclusion of methods for named entity translation.

>> Section 3

International Association for Machine Translation (IAMT) Archived 24 June 2010 at the Wayback Machine Machine Translation Archive Archived 1 April 2019 at the Wayback Machine by John Hutchins. An electronic repository (and bibliography) of articles, books and papers in the field of machine translation and computer-based translation technology Machine translation (computer-based translation) – Publications by John Hutchins (includes PDFs of several books on machine translation) Machine Translation and Minority Languages John Hutchins 1999 Archived 7 September 2007 at the Wayback Machine

>> Section 4

1990s and early 2000s By 1998, "for as little as $29.95" one could "buy a program for translating in one direction between English and a major European language of your choice" to run on a PC. MT on the web started with SYSTRAN offering free translation of small texts (1996) and then providing this via AltaVista Babelfish, which racked up 500,000 requests a day (1997). The second free translation service on the web was Lernout & Hauspie's GlobaLink. Atlantic Magazine wrote in 1998 that "Systran's Babelfish and GlobaLink's Comprende" handled "Don't bank on it" with a "competent performance." Franz Josef Och (the future head of Translation Development AT Google) won DARPA's speed MT competition (2003). More innovations during this time included MOSES, the open-source statistical MT engine (2007), a text/SMS translation service for mobiles in Japan (2008), and a mobile phone with built-in speech-to-speech translation functionality for English, Japanese and Chinese (2009). In 2012, Google announced that Google Translate translates roughly enough text to fill 1 million books in one day.

>> Section 5

A deep learning-based approach to MT, neural machine translation has made rapid progress in recent years. However, the current consensus is that the so-called human parity achieved is not real, being based wholly on limited domains, language pairs, and certain test benchmarks, i.e., it lacks statistical significance power. Translations by neural MT tools like DeepL Translator, which is thought to usually deliver the best machine translation results as of 2022, typically still need post-editing by a human. Instead of training specialized translation models on parallel datasets, one can also directly prompt generative large language models like GPT to translate a text. This approach is considered promising, but is still more resource-intensive than specialized translation models.

>> Section 6

Machine translation is the use of computational techniques to translate text or speech from one language to another, including the contextual, idiomatic, and pragmatic nuances of both languages. While some language models are capable of generating comprehensible results, machine translation tools remain limited by the complexity of language and emotion, often lacking depth and semantic precision. Its quality is influenced by linguistic, grammatical, tonal, and cultural differences, making it inadequate to replace real translators fully. Effective improvement in translation quality requires understanding of target society's customs and historical context, and human intervention and visual cues remain necessary in simultaneous interpretation. On the other hand, domain-specific customization, such as for technical documentation or official texts, can yield more stable results, and is commonly employed in multilingual websites and professional databases. Initial approaches were mostly rule-based or statistical in nature. However, these methods have since been superseded by neural machine translation and large language models.